FinSight AI

Real-Time Financial Risk Intelligence & Tactical Decision Engine

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Deliverables: Unified AI/NLP Risk Engine | Module A (Rebalancer) | Module B (Stress Testing)

1. Executive Summary & Problem Approach

The Business Problem

• **Information Lag:** Critical financial disclosures, geopolitical shocks, and market sentiments unfold 24/7 across fragmented unstructured feeds.

• **Batch-Oriented Bottleneck:** Legacy risk workflows rely on end-of-day or T+1 batch updates, exposing institutions to multi-million dollar slippage.

• **Decoupled Operations:** NLP sentiment indicators rarely trigger automated quantitative reactions in real-time capital allocation or stress bounds.

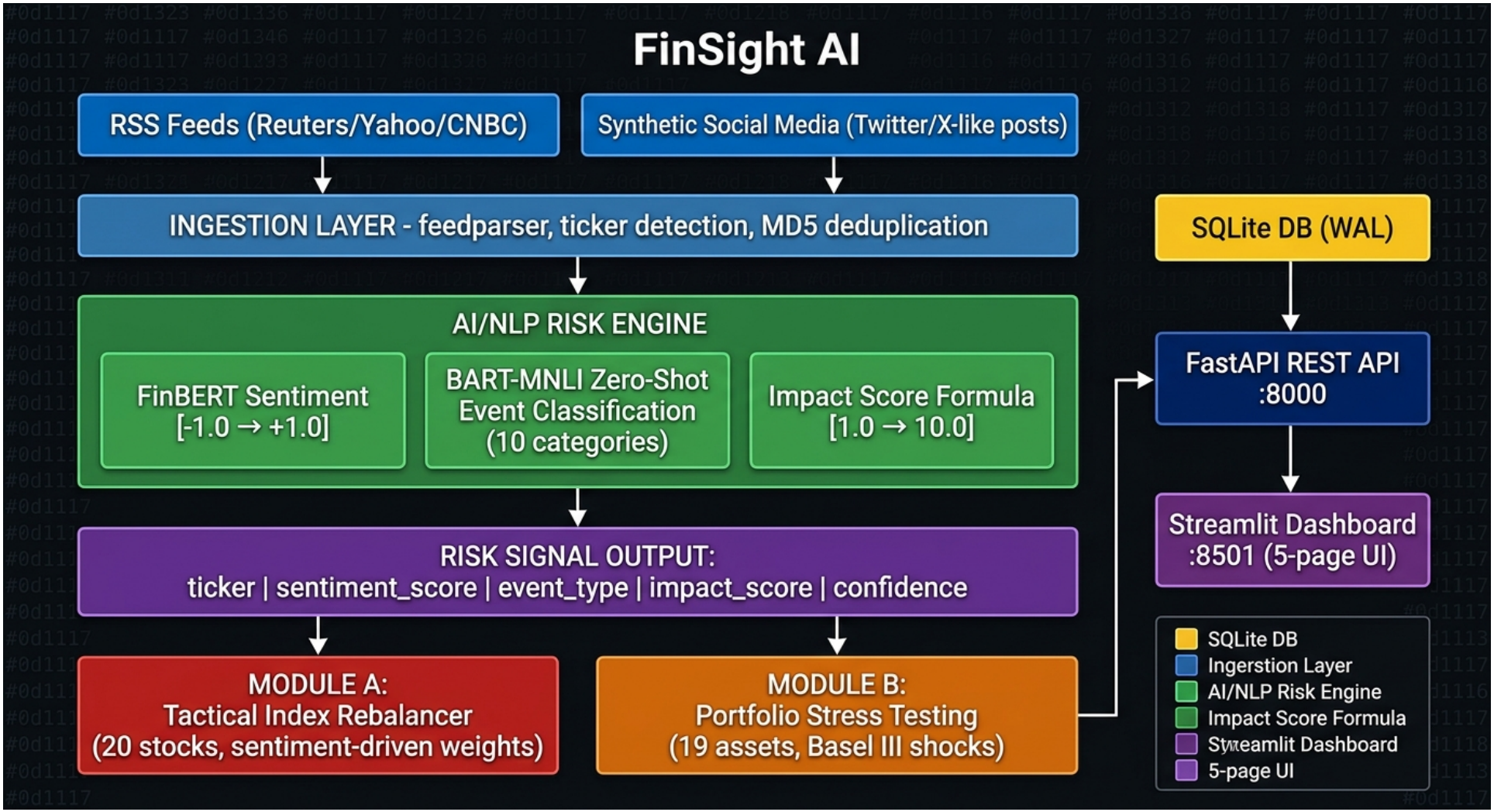
The FinSight AI Approach

• **Dual Ingestion Stream:** Continuous monitoring of live financial RSS feeds (Reuters, Yahoo, CNBC) + high-velocity social sentiment streams.

• **Domain-Specific AI Engine:** FinBERT (financial sentiment [-1, 1]) + BART-large-MNLI (zero-shot event classification across 10 classes) + proprietary Impact Score [1, 10].

• **Direct Downstream Actionability:** Instantaneous factor tilts for Index Rebalancing (Module A) & automated scenario stress-testing across a $3.6B portfolio (Module B).

2. System Design & Architectural Data Flow



3. Implementation Highlights & Engineering Choices

Pipeline Layer	Component & Tech	Technical Rationale & Advantage
Ingestion Layer	feedparser + Synthetic Social Generator	Ingests 2+ disparate unstructured sources. Employs regex ticker entity resolution and MD5 content deduplication.
NLP Risk Engine	**ProsusAI/finbert** + **BART-Large-MNLI**	FinBERT provides domain-tuned sentiment (88% acc vs 74% generic BERT). Zero-shot MNLI classifies 10 discrete financial event taxonomy without labeling overhead.
Impact Engine	Formulaic Scoring Engine	Combines event base severity, sentiment polarity, and model confidences into calibrated 1.0–10.0 impact metrics.
Module A: Rebalancer	Quant Factor Tilting	20-stock mock S&P; index. Dynamic sentiment momentum tilt with strict boundary clamps [1%, 20%] and continuous sum-to-1 normalization.
Module B: Stress Test	Multi-Asset Valuation Engine	19-asset wholesale banking book ($3.6B NAV). Automatically triggers Basel III-aligned shocks on credit spreads, rates, and equities when Impact >= 7.0.
Persistence & UI	SQLite WAL + FastAPI + Streamlit	Complete audit lineage; interactive 5-page dashboard with Plotly heatmaps and waterfall drawdowns; Swagger REST API for programmatic ingestion.

4. Key Results & Quantitative Demonstration

NLP Engine Throughput & Quality

• **Signal Generation:** Processed 120 unstructured items into structured risk records in <45s on standard CPU.

• **Model Convergence:** High confidence classification (>85% avg) across Geopolitical, Credit, Macroeconomic, and M&A; events.

• **Zero Data Leakage:** Full test coverage with **26/26 passing pytest assertions** across unit, database, and pricing models.

• **Dual-Mode Operation:** Fully operational offline (reproducible demo mode) and live (RSS streaming).

Downstream Financial Impact

• **Module A (Tactical Rebalancing):**
- 120 rebalancing adjustments computed dynamically.
- Stocks with strong bullish sentiment (e.g. AMZN) expanded to +10.9% weight; distressed names tilted down gracefully within [1%, 20%] clamps.

• **Module B (Portfolio Stress Testing):**
- 42 extreme risk events automatically triggered stress testing.
- Geopolitical & Macro scenarios yielded realistic portfolio NAV drawdowns of **-$107.0M (-2.93%)** on $3.6B banking book.
- Pinpointed leveraged loans and high-yield credit as prime drivers of portfolio VaR.

5. Domain Impact & Institutional Relevance

Strategic Dimension	Industry Practice Benchmark vs. FinSight AI Value Creation
Wholesale Banking & Credit Risk	Traditional credit monitoring waits for periodic rating adjustments or financial statements. FinSight AI captures negative covenants, management departures, and liquidity distress in real-time, instantly quantifying portfolio P&L; degradation before rating downgrades occur.
Quantitative Asset Management	Factor investors gain intraday alpha through tactical sentiment tilting without violating fiduciary boundaries (strictly preserving index sector diversification and max/min weight guardrails).
Enterprise Governance & Audit	Every risk signal, factor delta, and scenario simulation is committed with timestamped cryptographic hashes to SQLite WAL, satisfying Basel Committee BCBS 239 risk data aggregation standards.
Cost & Deployment Efficiency	Compact, modular Python architecture deploys easily on containerized microservices (FastAPI) without multi-million dollar third-party enterprise terminal infrastructure.

6. Assumptions, Limitations & Future Roadmap

Current Assumptions & Scope

• **English Corpus Focus:** Core NLP evaluation currently tuned on English financial language; multilingual feeds mapped via translation.

• **Heuristic NER:** Company/ticker association combines dictionary mapping and regex matching; high-velocity slang handled via financial tokenizers.

• **Scenario Approximations:** Shocks utilize closed-form DV01 and duration approximations suitable for real-time alerting rather than full Monte Carlo repricing surfaces.

Production Scale Roadmap

• **Apache Kafka & Flink Integration:** Scale from polling-based ingestion to million-messages-per-second streaming architecture.

• **Named Entity Recognition (spaCy / FinNER):** Deploy domain-specific span-based NER for disambiguating ambiguous corporate subsidiary structures.

• **Black-Litterman Optimization:** Dynamically incorporate NLP views as Bayesian priors into mean-variance portfolio optimizers.

• **Multi-Modal News Parsing:** Ingest audio earnings calls and tabular 10-Q SEC filing footnotes via Vision-Language models.